SCAN Foundations

PSYCH 511.001 Spring 2026

2026-01-12

Foundations of Social, Cognitive, and Affective Neuroscience (SCAN)

Instructor

Rick O. Gilmore, Ph.D. Professor of Psychology 114 Moore Building

+1 (814) 865-3664 rog1 AT-SYMBOL psu PERIOD edu Schedule an appointment Lab site: https://gilmore-lab.github.io

Where & When

467 Moore Thursdays, 1-4 PM

About the course

The first scientific psychologists were physiologists fascinated by the possibility of understanding the mind by studying the brain. In this course, we will explore the historical roots and contemporary challenges associated with the study of biological approaches to complex adaptive behavior. In doing so, we will read and examine critically primary source readings that discuss basic patterns and processes of brain structure and function. The goal is to provide students with a basic foundation of knowledge about the structures and functions of the nervous system that can provide the basis for future study.

This course is one of two required courses for the Specialization in Cognitive and Affective Neuroscience (SCAN).

Prerequisites

Undergraduate coursework in neuroscience or physiological psychology such as the equivalents of PSYCH 260 or BIO 469/470.

Week 01: Jan 12-16

Thu, Jan 15

Topic(s)

- Structure of the course
- Levels of analysis
- Causality in brain and behavior
- Does neuroscience need behavior? What does psychology need?

Read/Watch

- Required
  - Krakauer et al. (2017)
  - Parada and Rossi (2018)
  - Newcombe (2017)
- Recomended
  - Siddiqi et al. (2022)
  - Churchland and Sejnowski (1988)
  - Favela (2020)
  - Ross and Bassett (2024)
  - National Geographic (2014)

Materials

- Slides

Assignments

- *Assigned*: Exercise LVL • Levels/Terms
  - *Due*: 2026-01-22
  - Canvas dropbox

Week 02: Jan 19-23

Thu, Jan 22

Topics

- Neuroanatomy

Read/Watch

- Required
  - Leicester Medical School Radiology (2021) (Through ~ 8:20)
  - Neuroanatomy notes
- Recommended
  - Challenged (2025a)
  - Challenged (2025b)

Materials

- Slides
- Allen Brain Atlas.

Assignments

- *Due*: Exercise LVL • Levels/Terms
  - Canvas dropbox
- *Assigned*: Exercise ANAT • Neuroanatomy
  - *Due*: 2026-01-29
  - Canvas dropbox

Week 03: Jan 26-30

Thu, Jan 29

Topic(s)

- Action

Read/Watch

- Required
  - Shenoy, Sahani, and Churchland (2013)
  - Goulding, Bollu, and Büschges (2025)
- Recommended
  - *Hardwired for Song and Dance: The Neurobiology of Language and Movement* (2023)
  - Wolpert (2011)
  - (2021)

Materials

- Slides

Assignments

- *Due*: Exercise ANAT • Neuroanatomy
  - Canvas dropbox

Week 04: Feb 2-6

Thu, Feb 05

Topic(s)

- Perception

Read/Watch

- Required
  - Rosenberg et al. (2023)
  - Khalsa et al. (2018)
- Recommended
  - Neuroscience (2025)
  - CrashCourse (2015)

Assignment(s)

- *Assigned*: Final project proposal
  - *Due*: 2026-03-05
  - Canvas dropbox
- *Assigned*: Exercise PA • Perception & Action
  - *Due*: 2026-02-12.
  - Canvas dropbox

Week 05: Feb 9-13

Thu, Feb 12

Topic(s)

- Learning & memory

Read/Watch

- Required
  - Squire and Wixted (2011)
  - D'Esposito and Postle (2015)

Assignment(s)

- *Due*: Exercise PA • Perception & Action
  - Canvas dropbox

Week 06: Feb 16-20

Thu, Feb 19

Topic(s)

- Language

Read/Watch

- Required
  - Hickok and Poeppel (2007)
  - Bourguignon and Lo Bue (2025)
  - Hagoort and Indefrey (2014)
- Recommended
  - Bourguignon and Lo Bue (2025)
  - Zada et al. (2025)

Week 07: Feb 23-27

Thu, Feb 26

Topic(s)

- Emotion

Read/Watch

- Required
  - Malezieux, Klein, and Gogolla (2023)
  - Watabe-Uchida, Eshel, and Uchida (2017)
  - Siegel et al. (2018)
- Recommended
  - LeDoux (2000)
  - Barrett (2017)

Assignment(s)

- *Assigned*: Exercise EMO • Measuring emotion
  - *Due*: 2026-03-05
  - Canvas dropbox

Week 08: Mar 2-6

Thu, Mar 5

Topic(s)

- Sleep & circadian rhythms

Read/Watch

- Required
  - Nir and Tononi (2010)
  - Rasch and Born (2013)

Assignment(s)

- *Due*: Exercise EMO • Measuring emotion
  - Canvas dropbox
- *Due*: Final project proposal
  - Canvas dropbox
- *Assigned*: Exercise ZZ • Sleep & Circadian Rhythms
  - *Due*: 2026-03-19
  - Canvas dropbox

Spring Break: Mar 9-13

NO CLASS

Week 09: Mar 16-20

Thu, Mar 19

Topic(s)

- Evolution & development of the nervous system

Read/Watch

- Required
  - Sachkova, Modepalli, and Kittelmann (2025)
  - Stiles and Jernigan (2010)
- Optional
  - Castrillon et al. (2023)
  - Larsen et al. (2023)
  - Rakic (2009)
  - Cao, Huang, and He (2017)
  - Charvet and Finlay (2012)
  - Hofman (2014)
  - Blumberg and Adolph (2023)

Assignments

- *Due*: Exercise ZZ • Sleep & Circadian Rhythms
  - Canvas dropbox
- *Assigned*: Exercise EVO • Evolution
  - *Due*: 2026-03-26.
  - Canvas dropbox
- *Assigned*: Exercise DEV • Development
  - *Due*: 2026-04-02.
  - Canvas dropbox

Week 10: Mar 23-27

Thu, Mar 26

Topic(s)

- Cellular neuroscience I
  - Anatomy
  - Physiology
    - Resting potential

Read/Watch

- Cellular neuroscience notes | PDF |
- Optional
  - Won et al. (2025)
  - Zeng and Sanes (2017)
  - Oliveira et al. (2015)
  - Distéfano-Gagné et al. (2023)

Assignment(s)

- *Assigned*: Exercise PHYS • Neurophysiology I
  - *Due*: 2026-04-02
  - Canvas dropbox

Week 11: Mar 30 - Apr 3

Thu, Apr 02

Topics

- Cellular neuroscience II
  - Action potential
  - Synaptic transmission

Read/Watch

- Cellular neuroscience notes

Assignment(s)

- *Due*: Exercise DEV • Development
  - Canvas dropbox
- *Due*: Exercise PHYS • Neurophysiology I
  - Canvas dropbox
- *Assigned*: Exercise AP • Neurophysiology II
  - *Due*: 2026-04-09
  - Canvas dropbox

Week 12: Apr 6-10

Thu, Apr 09

Topic(s)

- Neurochemistry
  - Neurotransmitters
  - Hormones
- Neurocomputing

Read/Watch

- Neurochemistry notes
- Optional: Sarkar et al. (2016).

Assignment(s)

- *Due*: Exercise AP • Neurophysiology II
  - Canvas dropbox
- *Assigned*: Exercise CHEM • Neurochemistry
  - *Due*: 2026-04-16.
  - Canvas dropbox

Week 13: Apr 13-17

Thu, Apr 16

Topic(s)

- Disorder & disease I

Read/Watch

- Required
  - Howes, Bukala, and Beck (2023)
  - Volk et al. (2015)

Assignment(s)

- *Due*: Exercise CHEM • Neurochemistry
  - Canvas dropbox

Week 14: Apr 20-24

Thu, Apr 23

Topic(s)

- Disorder & disease II

Read/Watch

- Required
  - Moncrieff et al. (2022)
  - Namkung, Kim, and Sawa (2017)

Week 15: Apr 27 - May 1

Thu, Apr 30

Topics

- Frontiers in the biology of behavior
- Beethoven and the Cerebral Symphony

Finals: May 4-8

Thu, May 07

- Final Project write-up due
  - Canvas dropbox

References

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